

Class 17: Population Scale Analysis

Henry (A16354124)

Q13

```
# Read the file directly from the URL
url <- "https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG00000172057.6.txt"
data <- read.table(url)

# Inspect the first few rows to make sure it looks right
head(data)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
# 1. Sample Size for each genotype
cat("Sample Size for each genotype:\n")
```

Sample Size for each genotype:

```
table(data$geno)
```

A/A	A/G	G/G
108	233	121

```
# 2. Median Expression for each genotype
cat("Median Expression for each genotype:\n")
```

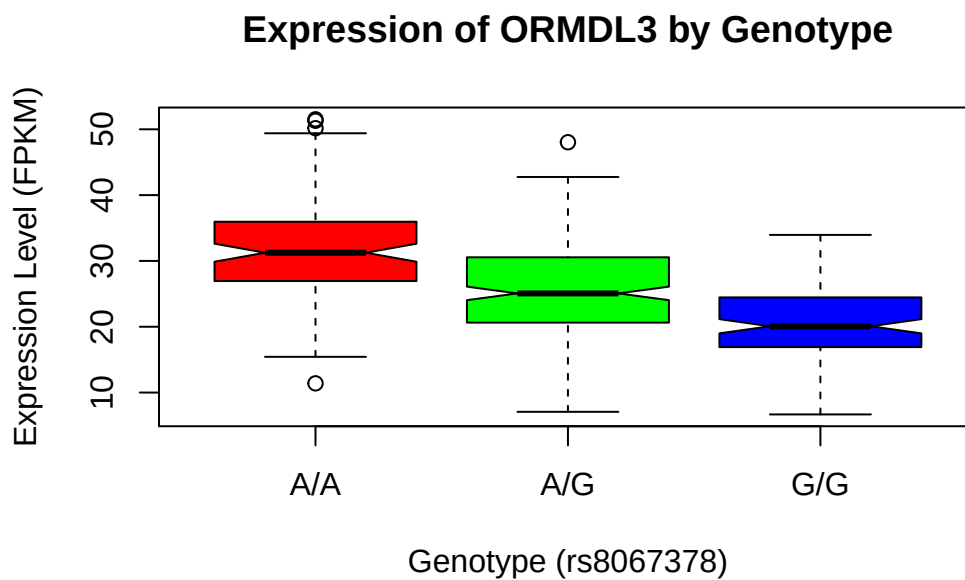
Median Expression for each genotype:

```
tapply(data$exp, data$geno, median)
```

```
      A/A      A/G      G/G
31.24847 25.06486 20.07363
```

Q14

```
# Boxplot of Expression by Genotype
boxplot(exp ~ geno, data = data,
        main = "Expression of ORMDL3 by Genotype",
        xlab = "Genotype (rs8067378)",
        ylab = "Expression Level (FPKM)",
        col = c("red", "green", "blue"),
        notch = TRUE)
```



```
# Perform One-Way ANOVA
# Formula: Expression depends on Genotype (exp ~ geno)
results_aov <- aov(exp ~ geno, data = data)

# Print the summary (Look for the "Pr(>F)" column, which is your P-value)
summary(results_aov)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
geno	2	7218	3609	75.66	<2e-16 ***
Residuals	459	21893	48		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1